

How Attachment Style Shapes Fear of Rejection and Friendship Formation Among BRAC University Undergraduates

Md. Monjurul Hasan Bhuiyan 1000054254

Khan Abdullah 1000054022

Azmyeen Nahian 1000054396

KM Mahir Labib 1000054525

Tasfia Hasan 24236091

Maysha Mahjabin 24226044

BRAC University

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Ms. Safina Binte Enayet

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Abstract

Attachment theory holds that early caregiving lays down a working model of whether other people can be relied on, and that this model governs how adults approach closeness. We asked how that model shows up in the friendships of undergraduates at BRAC University, and we asked it twice, once with numbers and once with talk. A survey of 103 students combined a four-paragraph self-categorisation of attachment style with continuous scales for attachment anxiety and avoidance, fear of negative evaluation, anticipated anxiety and expected acceptance across six concrete friendship bids, and loneliness. A focus group of nine students then worked through a vignette about a classmate who drafts a message asking to join a group and deletes it. Three results stand out. First, the paragraph a student pointed to predicted how many close friends they actually had, while the two continuous scales predicted nothing at all ($R^2 = .19$ against $R^2 = .01$; cross-validated $R^2 = .13$ against $-.06$). Second, how anxious a student expected to feel about making a friendly approach was unrelated to how well they expected it to go, in every one of the six situations we asked about. Third, in the focus group, the questions written specifically to draw out a personal instance produced the fewest: 6.7% of turns against 34.6% for questions that asked for nothing of the kind. Students at this campus recognise the pattern readily and describe it fluently. They do not claim it.

Keywords: attachment style, rejection sensitivity, loneliness, friendship formation, self-report validity, Bangladesh

Introduction

BY THE TIME a student walks into a first class at BRAC University they have already spent eighteen or twenty years learning what closeness costs. Bowlby (1969/1982) argued that the pattern of those early transactions is abstracted into an *internal working model*, a representation of whether others will respond that runs largely outside awareness. Ainsworth et al. (1978) sorted the resulting patterns into types; Main and Solomon (1990) added a fourth for children whose caregiver was both the danger and the only refuge from it. What concerns us is the residue. A model built at home does not stay at home. It shows up in who types a message and deletes it, and in who leaves the seat beside a friendly classmate empty.

Bartholomew and Horowitz (1991) carried this into adulthood by crossing a model of self against a model of others, producing four prototypes: secure, preoccupied, dismissing, and fearful. Each is a short paragraph, and respondents pick the one that fits. Downey and Feldman (1996) supplied the mechanism. People who anxiously expect rejection withdraw from the situations in which acceptance could occur, so the expectation makes itself true. Mikulincer and Shaver (2016) separate two routes through that machinery: anxious individuals *hyperactivate*, scanning for signs of exclusion, while avoidant individuals *deactivate*, treating needing people as a category error.

Operational definitions

Attachment style is used here in two distinct senses that the paper keeps apart on purpose. The *categorical* sense is the prototype a student selected from the four paragraphs. The *dimensional* sense is a pair of continuous scores, attachment anxiety and attachment avoidance, taken from short scales in the tradition of Fraley, Waller, et al. (2000). *Fear of negative evaluation* is apprehension about others' judgements, measured with straightforward-worded items following Rodebaugh et al. (2004). *Loneliness* is the perceived shortfall between wanted and achieved connection, which is not the same thing as being alone (Cacioppo & Hawkley, 2009). *Friendship formation* we treat behaviourally: the number of close friends held, and the number of conversations started with a stranger this semester.

The gap

The field has settled, largely, on dimensions. Fraley and Waller (1998) found no latent attachment taxon, Fraley et al. (2015) confirmed it, and MacCallum et al. (2002) showed what carving a continuum into groups costs in power and precision. But that literature concerns latent structure and how to score an instrument. Raby et al. (2021, p. 74) are careful that the evidence does not challenge the validity of those instruments as ways of collecting information. Whether categories or dimensions *predict better*, in the same students on the same afternoon, has been treated as settled by inference. We could not find a study that ran it.

Objectives, Rationale, and Hypotheses

We chose this topic because three of us had watched a friend go quiet across a semester, and none of us had a vocabulary for it beyond calling them shy. Objectives: (1) describe how students of each style talk about approaching people and being turned down; (2) test whether the fears that sit between style and behaviour are the ones students report; (3) compare the three insecure patterns rather than lumping them.

We expected insecure students to report more fear of negative evaluation and loneliness and to hold fewer close friends (H1); anxiety and avoidance to predict those outcomes dimensionally (H2); and anticipated anxiety about a friendly approach to track expected rejection (H3). H1 held. H2 and H3 did not, and their failure is what the paper is mostly about.

Methodology

Design and a departure from the proposal

Our approved proposal promised sixteen interviews and three focus groups. We ran one focus group and added a survey, because 73 of 103 respondents (70.9%) declined any follow-up contact, leaving too small a pool to fill sixteen slots in three weeks. The result is a convergent mixed-methods design, and the refusal rate turned out to be a finding rather than an obstacle.

Participants

Respondents were 103 BRAC University undergraduates ($M_{\text{age}} = 21.8$, $SD = 1.20$; 62 men, 39 women, 2 preferring not to say), recruited through course group chats and club pages between 16 and 18 August 2026. One further response was excluded because every demographic field was impossible, including an age of 10^8 . Computer Science and Engineering supplied 53 respondents, a limitation we return to. Focus group participants were nine students from EEE, CSE, and Biotechnology, in semesters five to nine.

Measures

Students read the four Bartholomew and Horowitz (1991) paragraphs, rated each, then chose one. Four-item scales measured attachment anxiety and avoidance. Six concrete bids, listed in Appendix F, were each rated twice, once for anticipated anxiety and once for expected acceptance, reproducing the architecture of Downey and Feldman's (1996) questionnaire. Six straightforward-worded items measured fear of negative evaluation, and three measured loneliness (Hughes et al., 2004). All six scales cleared $\alpha = .70$; full psychometrics are in Appendix C.

Focus group and ethics

The guide ran fourteen questions around a vignette about "Rafi" (Appendix F). Words such as *attachment* and *avoidant* were never spoken in the room. Consent was written and covered recording and quotation; participants could pass on anything; names became pseudonyms; everyone left with the contact card for the BRAC University Counselling and Wellness Centre. We describe patterns and diagnose nobody. Every value changed during cleaning is logged in Appendix B.

Results

Recognition predicts what report does not

The four styles differed as a set, Pillai's $V = .56$, $F(18, 285) = 3.67$, $p < .001$. Loneliness separated them most sharply ($\eta^2 = .26$), then conversations initiated (.20) and close friends held (.19). Secure students held 4.56 close friends against 1.93 for fearful students ($d = 1.06$), and the proportion with no close friend at all rose from 3.1% to 29.6%, trend $z = 2.88$, $p = .004$.

Attachment avoidance, measured continuously, did none of this work. It did not differ across

the four styles, $F(3, 99) = 1.36$, $p = .259$, and was unrelated to close friends held, $r = -.04$. Table 1 and Figure 1 give the comparison. For close friends held, the two dimensions explained 1.3% of variance and the prototype 19.4%. Under ten-fold cross-validation the dimensions scored $R^2 = -.06$, worse than predicting the sample mean, against $+.13$ for the prototype (bootstrapped difference $.181$, 95% CI $[.047, .316]$). The pattern reversed for fear of negative evaluation. Prototype choice predicts what a student's social life looks like; the scales predict what a student says they feel.

Anxiety is not a forecast

Within each of the six bids, a student's anticipated anxiety was uncorrelated with their expected acceptance (r from $-.004$ to $.147$, all $p > .13$). Aggregated, $r = .18$, $p = .076$, and equivalence testing could not exclude an effect as large as $.30$, so we claim independence cautiously. Between situations the relationship was strong and negative, Spearman $\rho = -.83$, $p = .042$: bids that frightened the room were the bids it expected to fail. The cheapest act was asking a classmate to explain something (anxiety 2.24, expected acceptance 4.32); the dearest was suggesting lunch (3.60, 3.44). Situations are ranked coherently. Students are not.

Which situations get avoided was likewise a property of the situation. Speaking in class was named by 64.1% and a cafeteria group by 60.2%, but group project meetings by only 13.6% (Cochran's $Q = 82.6$, $p < .001$). No single situation differed by attachment style (all $p > .38$).

The room would not say it

Across 74 coded turns and 3,581 words, 16.9% were first-person experiential, 38.0% first-person but hypothetical, and 45.1% generic. Three of the fourteen guide questions produced no usable transcript, and they were the three most exposing: childhood, being left out, and withheld disagreement. In 3,581 words there is one reference to family and one real person named. Question 11 asked what happens in the body before speaking to someone new; nobody answered.

The strongest result here is the one we did not expect. Questions written to extract a personal instance produced first-person disclosure in 3 of 45 turns (6.7%). Questions that asked for nothing of the kind produced it in 9 of 26 (34.6%), Fisher's exact $p = .006$, odds ratio 0.13. Disclosure did not vary by speaker ($p = .29$), so this is a property of the room rather than of a few reticent individuals. When it broke through, it came as recognition of the character, not description of the

self. One participant, asked nothing in particular, said: “I have been Rafi.”

Discussion with Reflection

The avoidance null is not a measurement failure, or not only one. Shaver and Mikulincer (2004) argue that people who act avoidantly may be unaware of it, or may deny it, and so rate themselves lower than they should on the very items meant to catch them. Dozier and Kobak (1992) found avoidant adults sweating through interviews they narrated calmly. There is a second, structural reason: Bartholomew and Horowitz split avoidance into dismissing and fearful, which load high on one bipolar dimension for opposite reasons, so a single score cannot separate them. Four-item scales attenuate everything, which is fair, but attenuation is symmetric and cannot explain why anxiety discriminated and avoidance did not in the same students.

Why should pointing at a paragraph work better? Niedenthal et al. (1985) showed that people make social decisions by matching themselves against a whole exemplar rather than decomposing the judgement into attributes. A prototype paragraph binds motivation, affect, and behaviour together, and Bowlby theorised working models as organisations rather than quantities. Recognising yourself in a configuration may preserve what scoring six symptoms discards. Participants chose the paragraph they had independently rated highest 81.6% of the time, so this is not random pointing.

The independence of anxiety and expectation fits Clark and Wells’s (1995) model, in which anticipatory anxiety is read off an internal image of how one appears rather than off evidence about the other person. Boothby et al. (2018) found the same dissociation: shyness predicted the size of their liking gap, rejection sensitivity did not. Downey and Feldman’s questionnaire multiplies anxiety by expected rejection and never reports how those components covary. In our data they barely do.

The focus group and the survey say the same thing in two registers. Seventy-one per cent declined a follow-up conversation, and refusal was flat across styles, $\chi^2(3) = 0.91, p = .82$. Reticence here is not the property of the insecure. It is the ambient condition. In the room, the group narrated Rafi’s inner sentence fluently and then declined to own it, and one of them located the whole problem inside himself: “It’s on us that we think or overthink. It’s our own psychological problem.” That sentence is the study in miniature. Fear becomes private property, and private property cannot be discussed.

Limitations

The sample skews to CSE and to men, the focus group had nine students and one session, and self-categorisation is known to be unstable across weeks (Baldwin & Fehr, 1995). Everything here is cross-sectional, so nothing licenses a causal claim about childhood. The pre-registered dissociation index we built failed (Appendix D), and we report it because it failed.

What we learned

We designed the guide believing that a well-built question extracts an answer. The opposite happened. Our best material came from Question 13, which the moderator softened on the spot into a question about expectations at admission, and which none of us thought was the important one. We had also assumed our job was to find the anxious students. The students who worried us by the end were the ones reporting the least distress and the fewest friends.

Conclusion and Recommendation

Among these 103 students, the prototype a person recognises themselves in predicted their friendships; two attachment scales did not. Anticipated anxiety about a friendly approach carried no information about how the approach would go. Which situations get avoided belongs to the campus, not the personality.

Two recommendations follow. Orientation and counselling outreach should use recognition rather than assessment, offering students descriptions to see themselves in instead of scales to score themselves on. And because the cheapest social act here is asking for academic help while the dearest is suggesting lunch, structured instrumental contact is the realistic lever: assigned study partners, staffed help desks, rotating seating. Students will accept an academic reason to talk long before a social one.

Psychology and Artificial Intelligence

Our instruments reached students through a browser rather than on paper. The survey ran as an online form, and the focus group guide was built as a web page that the moderator and the note-taker could both follow on their phones while the session ran. Building and circulating pages like that is cheap

now, and AI tools make it cheaper: they draft the markup, flag a question that reads ambiguously, and lay out a long item list so a participant can finish in under nine minutes.

The convenience matters for the psychology and not only for the logistics. Our clearest finding is that students recognise themselves in a description more readily than they rate themselves on a scale, and 71% of them declined to speak to us in person. A screen is easier to answer than a room. An instrument delivered online, and in time a conversational one that offers descriptions instead of items, reaches the students least likely to walk into a counselling office on their own.

The limits are the ordinary ones. Whatever a student types has to be stored carefully, and a form is not a diagnosis.

Tables and Figures

Table 1

Variance in each outcome explained by the two Likert dimensions, by the prototype the student chose, and by both

Outcome	R^2			Cross-validated R^2	
	Dimensions	Prototype	Both	Dimensions	Prototype
Close friends held	.013	.194	.194	-.059	.126
Conversations initiated	.007	.197	.198	-.054	.136
Loneliness	.148	.255	.340	.093	.188
Approach inhibition	.085	.073	.131	.023	-.022
Fear of negative evaluation	.141	.113	.195	.077	.032

Note. $n = 103$. Dimensions are the four-item attachment anxiety and avoidance scales. Prototype is the single four-category choice, entered as three dummy variables. Cross-validation is ten-fold, averaged over 20 repartitions; a negative value means the model predicts held-out cases worse than the sample mean does. Bold marks the better of the two predictors within each row. The prototype improved on the dimensions significantly for close friends held, $F(3, 97) = 7.21$, $p < .001$, for conversations initiated, $F(3, 97) = 7.72$, $p < .001$, and for loneliness, $F(3, 97) = 9.38$, $p < .001$, and not for the remaining two outcomes.

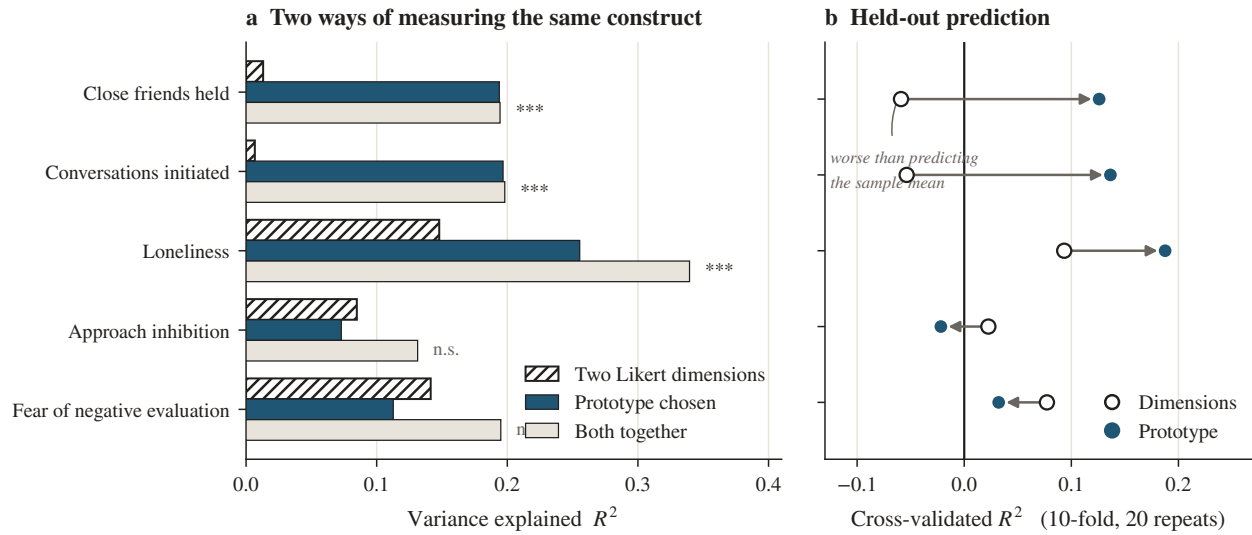


Figure 1

Recognition predicts behaviour; the scales predict reported feeling. Panel (a) gives variance explained in the full sample. Panel (b) gives held-out prediction, with the arrow running from the dimensional model to the prototype model. *** $p < .001$; n.s. = not significant.

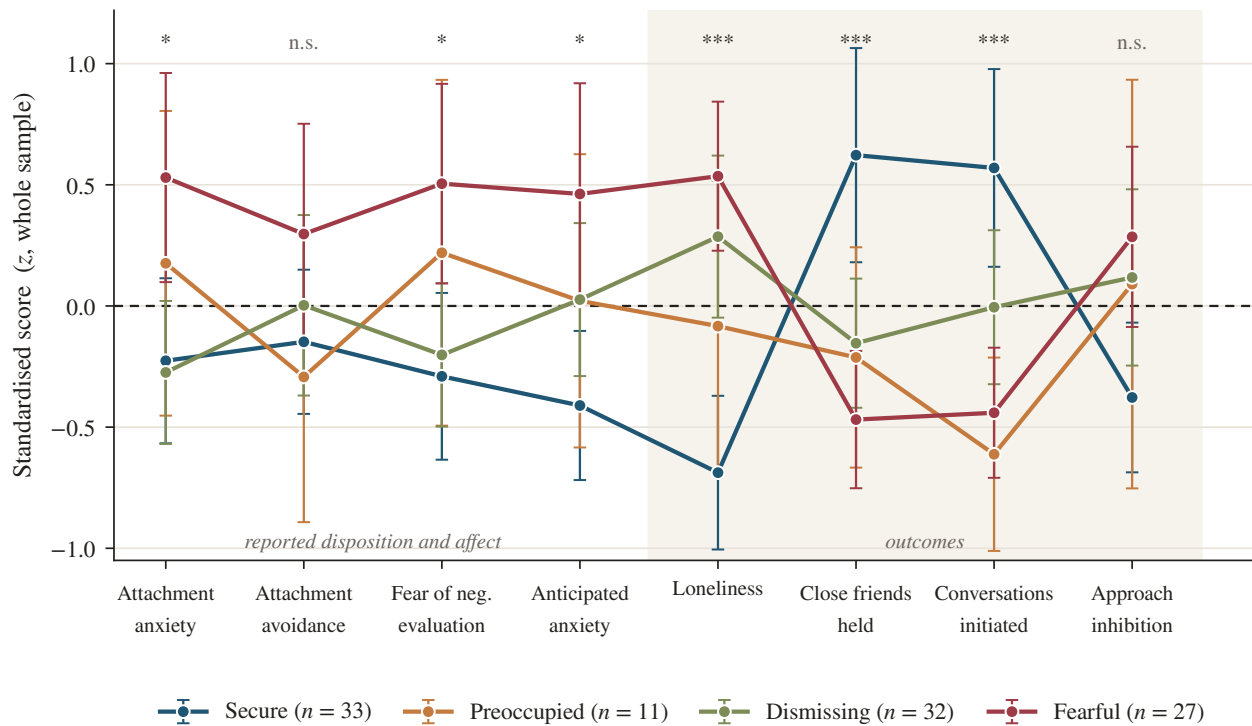
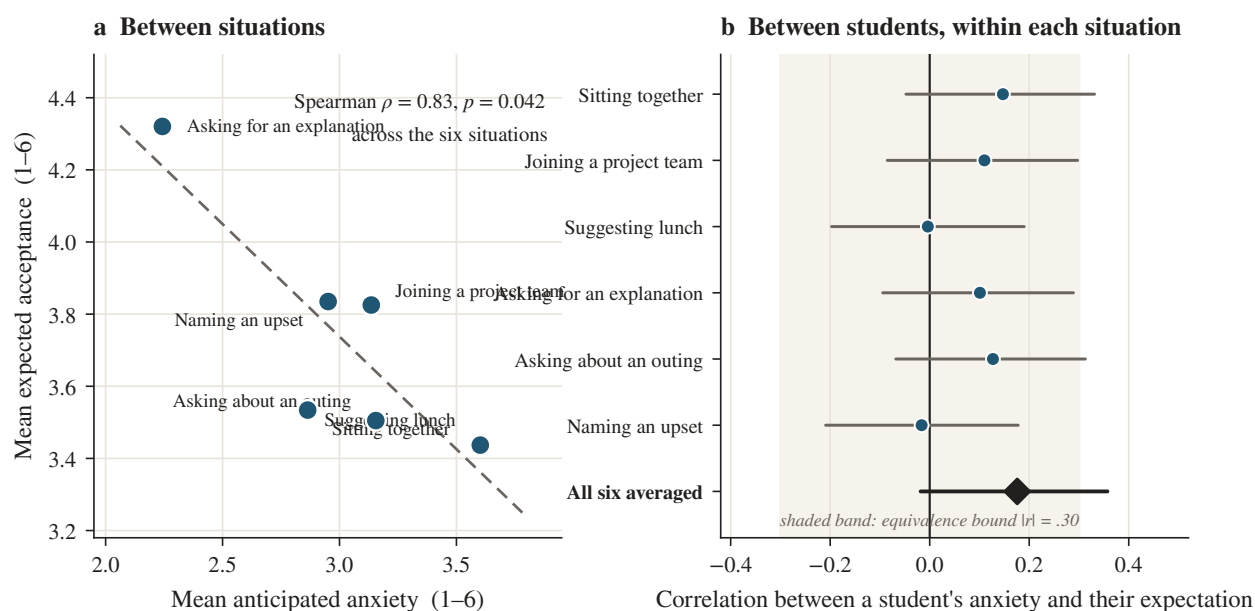
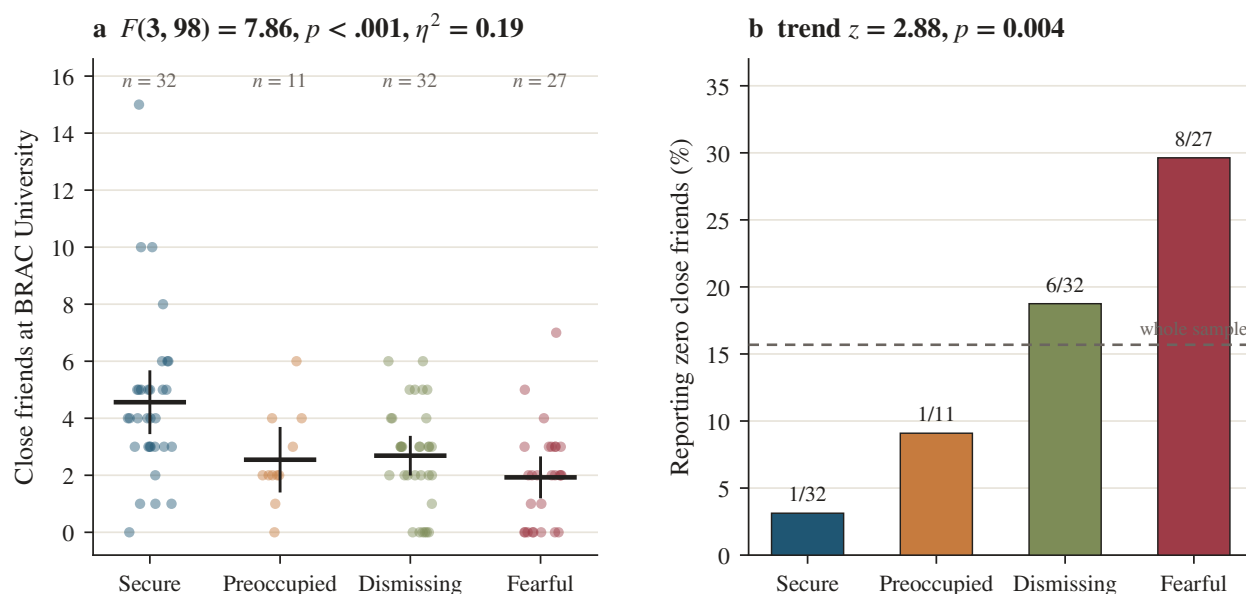


Figure 2

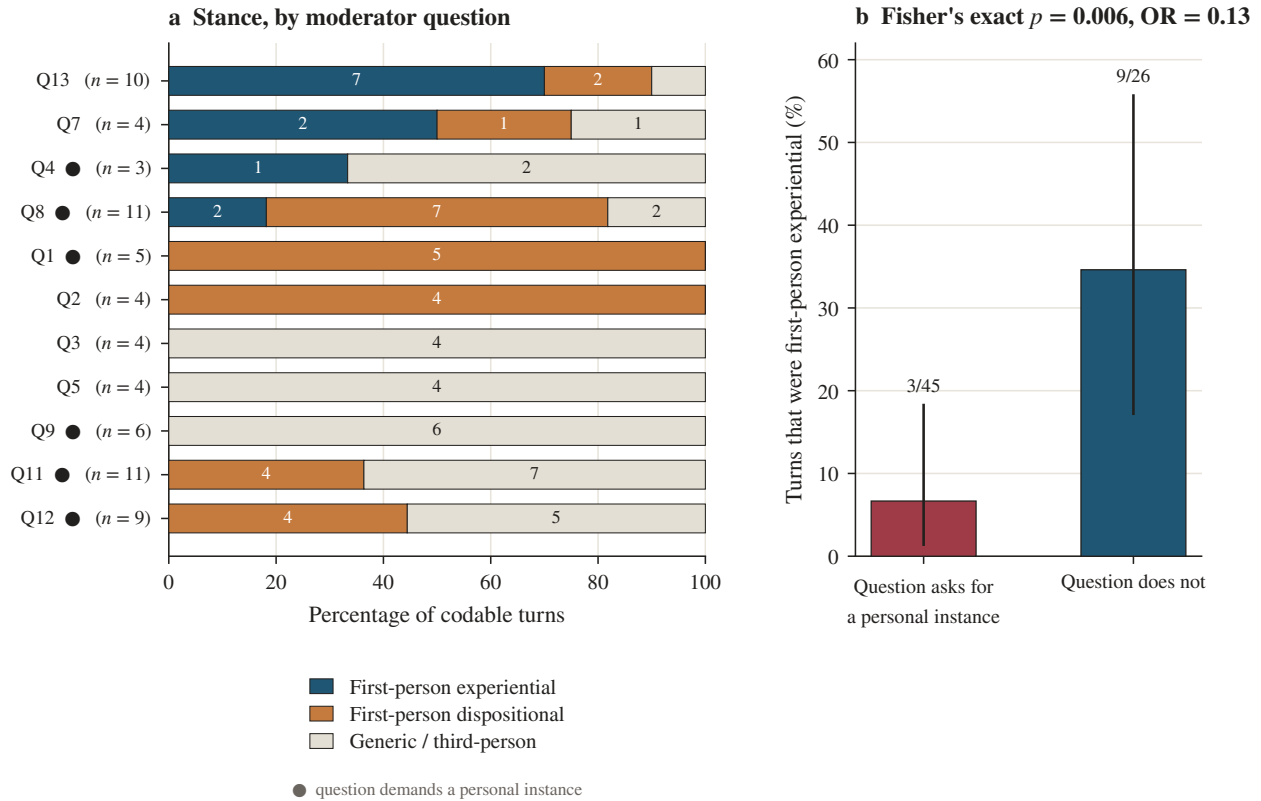
Standardised profile of the four self-selected styles across reported disposition and behavioural outcome. Bars are 95% confidence intervals. Asterisks mark one-way tests surviving Benjamini–Hochberg correction across the ten outcomes. Attachment avoidance, the second measure from the left, does not separate the groups.

**Figure 3**

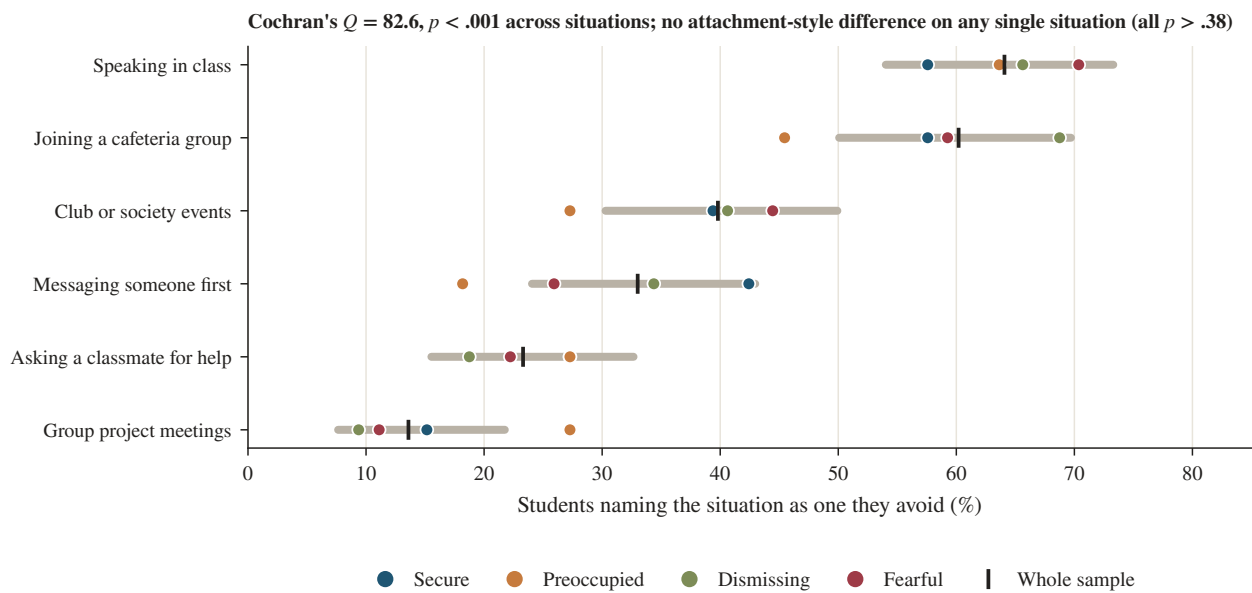
Anticipated anxiety and expected acceptance are strongly related across situations and unrelated across students. Panel (a) plots the six bids by their group means. Panel (b) gives the within-situation correlation between a student's anxiety and their own expectation, with 95% confidence intervals.

**Figure 4**

Close friends held, by style. Panel (a) shows every respondent, with the group mean and its 95% confidence interval. Panel (b) gives the proportion reporting no close friend at all, tested for monotone trend across the ordered styles.

**Figure 5**

Stance in the focus group. Panel (a) breaks each moderator question into first-person experiential, first-person dispositional, and generic turns. Panel (b) contrasts questions that demanded a personal instance with questions that did not. Bars are 95% confidence intervals.

**Figure 6**

Situations students say they avoid. The grey bar is the 95% confidence interval for the whole sample; coloured points give the four styles separately.

Implementation Plan

1. Implementation context

We would implement this at BRAC University itself, through the Counselling and Wellness Centre and the Office of Co-curricular Activities, targeting first and second semester undergraduates during Orientation and the first six weeks. The setting is appropriate for three reasons drawn from our own data. Nearly a third of respondents were in their first two years, the period our focus group identified as the only window in which approaching strangers still feels permissible. The open-credit system dissolves the cohort every semester, so peer groups have to be rebuilt repeatedly rather than once. And 15.5% of our sample reported no close friend at all, a figure high enough to justify a universal rather than a referral-based approach.

2. Connection to artificial intelligence

Our central result is that students recognise themselves in a description far more accurately than they rate themselves on a scale, and that they will not volunteer personal information when asked directly. Both point to the same technology: a conversational, retrieval-grounded chatbot delivered through the university portal that works by offering descriptions rather than administering items. Instead of asking “how often do you worry others will reject you,” it presents short scenarios of the kind we used and asks which one sounds familiar. A lightweight classifier over the resulting choices routes the student towards an appropriate resource. Natural-language processing over anonymised free-text answers could then surface the situations students actually avoid, updating the intervention each semester.

3. Implementation process

Semester one: secure Institutional Review Board approval, convene a working group of two counsellors, one data-protection officer, and four student representatives, and adapt the vignette bank. Semester two: pilot with two sections, roughly 80 students, measuring uptake and close-friend counts at weeks 1 and 12. Semester three: revise and extend to all incoming sections, and pair it with the structural measure our results actually support, namely assigned study partners and a staffed academic help desk, because asking for academic help was the cheapest social act we measured.

Resources needed are modest: existing portal infrastructure, one part-time developer, counsellor time, and no new hardware.

4. Potential benefits and challenges

The main benefit is reach without exposure. A student who declines a counselling appointment, as 71% of our respondents declined a follow-up conversation, may still answer a private screen at 2 a.m. The risks are serious and specific. Anything a student types is sensitive; storage must be minimised, encrypted, and never linked to academic records. A chatbot is not a clinician and must escalate rather than treat. Any predictive model trained on a CSE-heavy, English-medium sample will generalise poorly, and a false negative here means a struggling student is passed over.

5. Recommendations

Keep humans in the loop at every escalation. Publish the data policy in plain Bangla and English before launch. Validate any classifier on a fresh cohort before acting on it. Evaluate against behaviour, not satisfaction: close friends held at week 12 is the outcome that matters. And run it as opt-in, because a population this reticent will read compulsion as surveillance.

References

- Ainsworth, M. D. S., Blehar, M. C., Waters, E., & Wall, S. (1978). *Patterns of attachment: A psychological study of the Strange Situation*. Lawrence Erlbaum.
- Arnett, J. J. (2000). Emerging adulthood: A theory of development from the late teens through the twenties. *American Psychologist*, 55(5), 469–480. <https://doi.org/10.1037/0003-066X.55.5.469>
- Baldwin, M. W., & Fehr, B. (1995). On the instability of attachment style ratings. *Personal Relationships*, 2(3), 247–261. <https://doi.org/10.1111/j.1475-6811.1995.tb00090.x>
- Bartholomew, K., & Horowitz, L. M. (1991). Attachment styles among young adults: A test of a four-category model. *Journal of Personality and Social Psychology*, 61(2), 226–244. <https://doi.org/10.1037/0022-3514.61.2.226>
- Bartholomew, K., & Shaver, P. R. (1998). Methods of assessing adult attachment: Do they converge? In J. A. Simpson & W. S. Rholes (Eds.), *Attachment theory and close relationships* (pp. 25–45). Guilford Press.
- Baumeister, R. F., & Leary, M. R. (1995). The need to belong: Desire for interpersonal attachments as a fundamental human motivation. *Psychological Bulletin*, 117(3), 497–529. <https://doi.org/10.1037/0033-2909.117.3.497>
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- Bergkvist, L., & Rossiter, J. R. (2007). The predictive validity of multiple-item versus single-item measures of the same constructs. *Journal of Marketing Research*, 44(2), 175–184. <https://doi.org/10.1509/jmkr.44.2.175>
- Boothby, E. J., Cooney, G., Sandstrom, G. M., & Clark, M. S. (2018). The liking gap in conversations: Do people like us more than we think? *Psychological Science*, 29(11), 1742–1756. <https://doi.org/10.1177/0956797618783714>
- Bowlby, J. (1982). *Attachment and loss: Vol. 1. Attachment* (2nd ed.). Basic Books. (Original work published 1969)
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*,

- 3(2), 77–101. <https://doi.org/10.1191/1478088706qp063oa>
- Braun, V., & Clarke, V. (2021). One size fits all? What counts as quality practice in (reflexive) thematic analysis? *Qualitative Research in Psychology*, 18(3), 328–352. <https://doi.org/10.1080/14780887.2020.1769238>
- Brennan, K. A., Clark, C. L., & Shaver, P. R. (1998). Self-report measurement of adult attachment: An integrative overview. In J. A. Simpson & W. S. Rholes (Eds.), *Attachment theory and close relationships* (pp. 46–76). Guilford Press.
- Buote, V. M., Pancer, S. M., Pratt, M. W., Adams, G., Birnie-Lefcovitch, S., Polivy, J., & Wintre, M. G. (2007). The importance of friends: Friendship and adjustment among 1st-year university students. *Journal of Adolescent Research*, 22(6), 665–689. <https://doi.org/10.1177/0743558407306344>
- Cacioppo, J. T., & Hawkley, L. C. (2009). Perceived social isolation and cognition. *Trends in Cognitive Sciences*, 13(10), 447–454. <https://doi.org/10.1016/j.tics.2009.06.005>
- Clark, D. M., & Wells, A. (1995). A cognitive model of social phobia. In R. G. Heimberg, M. R. Liebowitz, D. A. Hope, & F. R. Schneier (Eds.), *Social phobia: Diagnosis, assessment, and treatment* (pp. 69–93). Guilford Press.
- Cortina, J. M. (1993). What is coefficient alpha? An examination of theory and applications. *Journal of Applied Psychology*, 78(1), 98–104. <https://doi.org/10.1037/0021-9010.78.1.98>
- Cozzarelli, C., Karafa, J. A., Collins, N. L., & Tagler, M. J. (2003). Stability and change in adult attachment styles. *Journal of Social and Clinical Psychology*, 22(3), 315–346. <https://doi.org/10.1521/jscp.22.3.315.22888>
- Creswell, J. W., & Plano Clark, V. L. (2018). *Designing and conducting mixed methods research* (3rd ed.). SAGE.
- Downey, G., & Feldman, S. I. (1996). Implications of rejection sensitivity for intimate relationships. *Journal of Personality and Social Psychology*, 70(6), 1327–1343. <https://doi.org/10.1037/0022-3514.70.6.1327>
- Downey, G., Freitas, A. L., Michaelis, B., & Khouri, H. (1998). The self-fulfilling prophecy in close relationships: Rejection sensitivity and rejection by romantic partners. *Journal of Personality and Social Psychology*, 75(2), 545–560. <https://doi.org/10.1037/0022-3514.75.2.545>
- Dozier, M., & Kobak, R. R. (1992). Psychophysiology in attachment interviews: Converging evidence

- for deactivating strategies. *Child Development*, 63(6), 1473–1480. <https://doi.org/10.2307/1131569>
- Epley, N., & Schroeder, J. (2014). Mistakenly seeking solitude. *Journal of Experimental Psychology: General*, 143(5), 1980–1999. <https://doi.org/10.1037/a0037323>
- Festinger, L., Schachter, S., & Back, K. (1950). *Social pressures in informal groups: A study of human factors in housing*. Harper.
- Finch, J. (1987). The vignette technique in survey research. *Sociology*, 21(1), 105–114. <https://doi.org/10.1177/0038038587021001008>
- Fraley, R. C., Garner, J. P., & Shaver, P. R. (2000). Adult attachment and the defensive regulation of attention and memory: Examining the role of preemptive and postemptive defensive processes. *Journal of Personality and Social Psychology*, 79(5), 816–826. <https://doi.org/10.1037/0022-3514.79.5.816>
- Fraley, R. C., Hudson, N. W., Heffernan, M. E., & Segal, N. (2015). Are adult attachment styles categorical or dimensional? A taxometric analysis of general and relationship-specific attachment orientations. *Journal of Personality and Social Psychology*, 109(2), 354–368. <https://doi.org/10.1037/pspp0000027>
- Fraley, R. C., & Waller, N. G. (1998). Adult attachment patterns: A test of the typological model. In J. A. Simpson & W. S. Rholes (Eds.), *Attachment theory and close relationships* (pp. 77–114). Guilford Press.
- Fraley, R. C., Waller, N. G., & Brennan, K. A. (2000). An item response theory analysis of self-report measures of adult attachment. *Journal of Personality and Social Psychology*, 78(2), 350–365. <https://doi.org/10.1037/0022-3514.78.2.350>
- Games, P. A., & Howell, J. F. (1976). Pairwise multiple comparison procedures with unequal n 's and/or variances. *Journal of Educational Statistics*, 1(2), 113–125. <https://doi.org/10.3102/10769986001002113>
- Griffin, D. W., & Bartholomew, K. (1994). Models of the self and other: Fundamental dimensions underlying measures of adult attachment. *Journal of Personality and Social Psychology*, 67(3), 430–445. <https://doi.org/10.1037/0022-3514.67.3.430>
- Hall, J. A. (2019). How many hours does it take to make a friend? *Journal of Social and Personal*

- Relationships*, 36(4), 1278–1296. <https://doi.org/10.1177/0265407518761225>
- Henrich, J., Heine, S. J., & Norenzayan, A. (2010). The weirdest people in the world? *Behavioral and Brain Sciences*, 33(2–3), 61–83. <https://doi.org/10.1017/S0140525X0999152X>
- Hubert, L., & Arabie, P. (1985). Comparing partitions. *Journal of Classification*, 2(1), 193–218. <https://doi.org/10.1007/BF01908075>
- Hughes, M. E., Waite, L. J., Hawkley, L. C., & Cacioppo, J. T. (2004). A short scale for measuring loneliness in large surveys: Results from two population-based studies. *Research on Aging*, 26(6), 655–672. <https://doi.org/10.1177/0164027504268574>
- Innamorati, M., Balsamo, M., Fairfield, B., Fabbriatore, M., Tamburello, A., & Saggino, A. (2014). Construct validity and reliability of the adult rejection sensitivity questionnaire: A comparison of three factor models. *Depression Research and Treatment*, 2014, 972424. <https://doi.org/10.1155/2014/972424>
- Leary, M. R. (1983). A brief version of the Fear of Negative Evaluation Scale. *Personality and Social Psychology Bulletin*, 9(3), 371–375. <https://doi.org/10.1177/0146167283093007>
- MacCallum, R. C., Zhang, S., Preacher, K. J., & Rucker, D. D. (2002). On the practice of dichotomization of quantitative variables. *Psychological Methods*, 7(1), 19–40. <https://doi.org/10.1037/1082-989X.7.1.19>
- Main, M., & Solomon, J. (1990). Procedures for identifying infants as disorganized/disoriented during the Ainsworth Strange Situation. In M. T. Greenberg, D. Cicchetti, & E. M. Cummings (Eds.), *Attachment in the preschool years: Theory, research, and intervention* (pp. 121–160). University of Chicago Press.
- Maliha, Z. T., Nahar, S. N., Mitra, D. K., Kakoly, N. S., Koly, K. N., Talukder, M. H. K., Ahmed, H. U., Gupta, R. D., & Hasan, M. T. (2025). Exploring loneliness: Prevalence and determinants among medical students in Bangladesh. *PLOS Mental Health*, 2(8), e0000379. <https://doi.org/10.1371/journal.pmen.0000379>
- Mikulincer, M., & Orbach, I. (1995). Attachment styles and repressive defensiveness: The accessibility and architecture of affective memories. *Journal of Personality and Social Psychology*, 68(5), 917–925. <https://doi.org/10.1037/0022-3514.68.5.917>
- Mikulincer, M., & Shaver, P. R. (2016). *Attachment in adulthood: Structure, dynamics, and change* (2nd ed.).

Guilford Press.

- Mischel, W., & Shoda, Y. (1995). A cognitive-affective system theory of personality. *Psychological Review*, 102(2), 246–268. <https://doi.org/10.1037/0033-295X.102.2.246>
- Niedenthal, P. M., Cantor, N., & Kihlstrom, J. F. (1985). Prototype matching: A strategy for social decision making. *Journal of Personality and Social Psychology*, 48(3), 575–584. <https://doi.org/10.1037/0022-3514.48.3.575>
- Raby, K. L., Fraley, R. C., & Roisman, G. I. (2021). Categorical or dimensional measures of attachment? Insights from factor-analytic and taxometric research. In R. A. Thompson, J. A. Simpson, & L. J. Berlin (Eds.), *Attachment: The fundamental questions* (pp. 70–77). Guilford Press.
- Rapee, R. M., & Heimberg, R. G. (1997). A cognitive-behavioral model of anxiety in social phobia. *Behaviour Research and Therapy*, 35(8), 741–756. [https://doi.org/10.1016/S0005-7967\(97\)00022-3](https://doi.org/10.1016/S0005-7967(97)00022-3)
- Rodebaugh, T. L., Woods, C. M., Thissen, D. M., Heimberg, R. G., Chambless, D. L., & Rapee, R. M. (2004). More information from fewer questions: The factor structure and item properties of the original and Brief Fear of Negative Evaluation Scale. *Psychological Assessment*, 16(2), 169–181. <https://doi.org/10.1037/1040-3590.16.2.169>
- Roisman, G. I. (2009). Adult attachment: Toward a rapprochement of methodological cultures. *Current Directions in Psychological Science*, 18(2), 122–126. <https://doi.org/10.1111/j.1467-8721.2009.01621.x>
- Russell, D. W. (1996). UCLA Loneliness Scale (Version 3): Reliability, validity, and factor structure. *Journal of Personality Assessment*, 66(1), 20–40. https://doi.org/10.1207/s15327752jpa6601_2
- Sandstrom, G. M., & Boothby, E. J. (2021). Why do people avoid talking to strangers? A mini meta-analysis of predicted fears and actual experiences talking to a stranger. *Self and Identity*, 20(1), 47–71. <https://doi.org/10.1080/15298868.2020.1816568>
- Sandstrom, G. M., & Dunn, E. W. (2014). Social interactions and well-being: The surprising power of weak ties. *Personality and Social Psychology Bulletin*, 40(7), 910–922. <https://doi.org/10.1177/0146167214529799>
- Scharfe, E., & Bartholomew, K. (1994). Reliability and stability of adult attachment patterns. *Personal Relationships*, 1(1), 23–43. <https://doi.org/10.1111/j.1475-6811.1994.tb00053.x>
- Shaver, P. R., & Mikulincer, M. (2004). What do self-report attachment measures assess? In W. S. Rholes &

- J. A. Simpson (Eds.), *Adult attachment: Theory, research, and clinical implications* (pp. 17–54). Guilford Press.
- Tinto, V. (1975). Dropout from higher education: A theoretical synthesis of recent research. *Review of Educational Research*, 45(1), 89–125. <https://doi.org/10.3102/00346543045001089>
- Wanous, J. P., Reichers, A. E., & Hudy, M. J. (1997). Overall job satisfaction: How good are single-item measures? *Journal of Applied Psychology*, 82(2), 247–252. <https://doi.org/10.1037/0021-9010.82.2.247>
- Watson, D., & Friend, R. (1969). Measurement of social-evaluative anxiety. *Journal of Consulting and Clinical Psychology*, 33(4), 448–457. <https://doi.org/10.1037/h0027806>
- Zajonc, R. B. (1968). Attitudinal effects of mere exposure. *Journal of Personality and Social Psychology*, 9(2, Pt. 2), 1–27. <https://doi.org/10.1037/h0025848>

Appendix A

Extended Literature Review

The main report compresses the literature into a page because of the word limit. This appendix gives the argument at proper length, organised by the three claims the study makes.

A.1 Attachment was typological by accident, and the field noticed

Attachment theory began as an account of organised behavioural strategies rather than of continuous traits. Bowlby (1969/1982) proposed that repeated transactions with a caregiver are abstracted into an internal working model that steers expectation, appraisal, and behaviour in later relationships. Because working models were conceived as organisations rather than as quantities, the first operationalisation was inevitably a sorting exercise. Ainsworth et al. (1978) rated Strange Situation behaviour on continuous scales but used those ratings to place infants into three classes, and Main and Solomon (1990) added a fourth for infants whose behaviour showed no coherent strategy at all. As Raby et al. (2021) point out, this sorting practice then served as the template for the adult measures built in the 1980s. The categorical apparatus was a methodological inheritance rather than a tested claim about latent structure.

Bartholomew and Horowitz (1991) reorganised the adult typology by crossing a model of self against a model of others, yielding four prototypes. The important move, for our purposes, was splitting avoidance in two. Dismissing avoidance pairs a positive self-model with a negative other-model; fearful avoidance pairs two negative models. Both look avoidant from outside, and both score high on a single bipolar avoidance scale, but they arrive there for opposite reasons.

The psychometric weaknesses of the resulting Relationship Questionnaire are well documented. Griffin and Bartholomew (1994) showed the single forced choice is a low-information indicator and recommended aggregating prototype ratings instead. Scharfe and Bartholomew (1994) found only moderate stability across eight months. Baldwin and Fehr (1995) found roughly 30% of respondents changed their selected category over intervals as short as a week, and Cozzarelli et al. (2003) found 46% of 442 women changed across two years. We take this seriously, and it cuts in an unexpected direction for our own results: a measure this noisy should be crippled by attenuation, yet it still produced $\Delta R^2 = .18$ and $.19$. Whatever signal it carries is presumably larger than what

we observed.

The decisive challenge to categories came from taxometrics. Fraley and Waller (1998) applied Meehl's procedures to 639 young adults and found no latent taxon. Brennan et al. (1998) recovered two broad orthogonal dimensions from the whole corpus of English-language attachment items, and Fraley, Waller, et al. (2000) rebuilt the item set with item response theory. Fraley et al. (2015) revisited the question with less subjective methods and concluded that individual differences appear more consistent with a dimensional than a categorical model. Alongside this, MacCallum et al. (2002) showed that dichotomising a genuinely continuous variable costs power, biases estimates, and inflates error rates in both directions.

What is easy to miss is how narrow the dimensionalist claim actually is. Raby et al. (2021, p. 74) state that the latent-structure evidence does not challenge the validity of the Strange Situation, the Adult Attachment Interview, or self-report questionnaires as instruments for collecting information about attachment; the argument is about how to score them. They further grant that categorical systems have been heuristically valuable. The consensus, in other words, is about latent distribution and about scoring efficiency. It is not, and has never been, a demonstration that categories predict worse. Roisman (2009) called for a rapprochement between methodological cultures, and Bartholomew and Shaver (1998) found convergence between attachment methods to be modest. We searched for a study running the head-to-head predictive comparison in a single sample and did not find one.

A.2 Why avoidance in particular should resist self-report

There is a specific and well-grounded reason to expect a continuous avoidance scale to underperform. Mikulincer and Shaver (2016) describe two secondary attachment strategies. Hyperactivation amplifies and broadcasts distress; deactivation suppresses attachment needs and maintains compulsory self-reliance. These have asymmetric consequences for a questionnaire. A hyperactivating student is, in effect, motivated to endorse anxiety items. A deactivating student is motivated to deny avoidance items. Shaver and Mikulincer (2004) put it plainly: people who behave avoidantly may be unaware of their avoidance, or may defensively deny their detachment, and so rate themselves lower than they should on the very items designed to detect it.

The supporting evidence does not rely on self-report. Dozier and Kobak (1992) found

adults using deactivating strategies showed elevated skin conductance while discussing separation and rejection, contradicting their verbal composure. Mikulincer and Orbach (1995) documented repressive defensiveness in the accessibility of avoidant individuals' affective memories. Fraley, Garner, et al. (2000) showed avoidant adults encode less attachment-relevant information in the first place, a defence operating before the material ever reaches the level at which a Likert item could interrogate it.

The prediction is therefore asymmetric and precise. A direct avoidance scale should discriminate poorly among self-nominated styles while a direct anxiety scale should discriminate acceptably. This is exactly the pattern in our data ($F = 1.36$, $p = .259$ for avoidance; $F = 4.41$, $p = .006$ for anxiety), which is why we treat the avoidance null as a predicted signature rather than as noise. Note that this and the Bartholomew structural argument are independent and mutually reinforcing: defensive under-reporting depresses the scores, and the collapse of two opposite avoidance types onto one dimension scrambles what remains.

A.3 Why pointing at a paragraph might work

Niedenthal et al. (1985) showed that people routinely make consequential social decisions by prototype matching, comparing a holistic self-image against an integrated exemplar rather than decomposing the judgement into attributes. A prototype paragraph presents motivation, affect, behaviour, and self-concept bound together. Recognising yourself in it is a judgement about a configuration. Endorsing "I worry that people close to me will leave me" is a judgement about the frequency of one symptom. Where the construct is genuinely configural, and Bowlby theorised working models as organisations, the configural judgement may retain information that item-level decomposition throws away.

This sits within a broader psychometric literature showing that single items are not automatically inferior. Wanous et al. (1997) meta-analysed 28 correlations across 7,682 people and found single-item job satisfaction measures correlated .67 with multi-item scales after correcting for reliability. Bergkvist and Rossiter (2007) found no predictive-validity difference for concrete constructs. None of this says single items are better. It says the presumption of multi-item superiority is an empirical question rather than a corollary of internal consistency, and internal consistency is precisely what our four-item scales have and our single choice does not.

Any study using four-item scales must confront the length problem directly. Coefficient alpha is a function of test length as well as of inter-item correlation (Cortina, 1993), and the well-validated short-form ECR needed twelve items. Our scales will attenuate observed correlations relative to the 36-item ECR-R. The defence is internal rather than rhetorical: attenuation is symmetric across both dimensions, so it cannot explain why anxiety discriminated and avoidance did not among the same participants in the same sitting, nor why a single unreliable choice added substantial incremental variance over both.

A.4 Rejection sensitivity, and the component that nobody reports

Downey and Feldman (1996) defined rejection sensitivity as the disposition to anxiously expect, readily perceive, and intensely overreact to rejection. Downey et al. (1998) then closed the loop empirically, showing that rejection-sensitive individuals behave in ways that elicit genuine rejection. The construct is explicitly a cognitive-affective processing disposition in the sense of Mischel and Shoda (1995), generating if–then situation–behaviour signatures rather than uniform output, which matters for interpreting our situation-dominant results.

The measurement detail is what our study speaks to. The Rejection Sensitivity Questionnaire presents eighteen situations and asks two questions about each on six-point scales: how concerned or anxious the respondent would be, and how likely the other person is to respond acceptingly. The two are then combined multiplicatively. Our six friendship bids reproduce that architecture deliberately. The original validation paper reports the reliability and factor structure of the composite but does not report the covariance between the two components it multiplies. Innamorati et al. (2014), working with 774 adults, found the questionnaire best represented by a bifactor model with two separable group factors, expectancy and anxiety, with different correlates, and only 44% of total-score variance attributable to common factors. Our within-situation correlations of essentially zero are a direct, situation-level demonstration of the same separability.

Why the two should come apart is explained by the cognitive models of social anxiety. Clark and Wells (1995) and Rapee and Heimberg (1997) both hold that socially anxious individuals construct and monitor an internal representation of how they appear, generated from interoceptive and self-focused information rather than from external feedback. Anticipatory anxiety is a readout of that internal image. Expected acceptance is a judgement about another person. Different inputs,

no necessary correlation. This framing matters for how we state the result: our data show anxiety is not a forecast, which is not the same as showing anxiety is a mistake.

Convergent evidence comes from the misprediction literature. Boothby et al. (2018) documented the liking gap, and two details are directly relevant. Shyness moderated its size, but rejection sensitivity did not ($p = .64$), nor did self-esteem or narcissism, an independent replication of exactly the dissociation we observe. And in their Study 5, following 102 first-year students across five waves, a \$100 accuracy incentive did not shift estimates, so the misprediction is sincerely held rather than performed modesty. Epley and Schroeder (2014), Sandstrom and Boothby (2021), and Sandstrom and Dunn (2014) all report the same asymmetry between predicted and experienced social contact.

A.5 Opportunity, and the friendship arithmetic

Friendship formation is driven by opportunity as much as by disposition. Festinger et al. (1950) showed proximity and functional distance predicted friendship in student housing; Zajonc (1968) showed repeated exposure alone increases liking. Hall (2019) quantified the dosage across a survey of recent movers and a three-wave study of 112 first-year students: roughly 40 to 60 hours together to move from acquaintance to casual friend, 80 to 100 to friend, and 200 or more to close friend, with leisure time counting far more than time spent working together.

Set that against our bid-cost ordering and a mechanism appears that Hall's hour-counting does not address. The cheapest social act we measured was instrumental and academic, asking a classmate to explain something. The dearest was relational, suggesting lunch. Students at this campus preferentially accrue exactly the kind of hours that Hall's data say do not build friendships, and avoid the kind that do. Tinto (1975) and Buote et al. (2007) place this in the higher-education context, and Arnett (2000) explains why the timing bites: emerging adulthood concentrates identity-relevant peer feedback at precisely the moment the existing network is dismantled.

A.6 Culture, disclosure, and the limits of a focus group

Two literatures bear on why our room would not speak in the first person. The first is methodological. Focus groups are known to suppress sensitive personal disclosure relative to individual interviews, because the audience for any admission is a set of peers who will still be on campus next week. This

is the standard rationale for vignette methods: Finch (1987) argued that vignettes let participants comment on a situation at one remove, which lowers the cost of engaging with material they would not volunteer about themselves. Our data suggest the technique works as advertised and then stops there. The vignette gave the room a shared object and a fluent vocabulary. It did not convert either into self-disclosure, and the one moment it did (“I have been Rafi”) was recognition of the character rather than description of the self.

The second is cultural, and here we are more careful than we would like to be. Henrich et al. (2010) established that the psychological literature is built disproportionately on Western, educated, industrialised, rich, and democratic samples, and that generalisation beyond them cannot be assumed. Bangladesh is a collectivist setting in which family difficulty is conventionally not discussed outside the family, and it would be easy to write a confident paragraph attributing our silence to that. We are not able to do so responsibly. We did not measure cultural orientation, we have no comparison sample, and our refusal rate was flat across attachment styles, which tells us the reticence is general but not why. Maliha et al. (2025) report elevated loneliness among Bangladeshi medical students, which is the closest local benchmark we found, and it concerns a different population. The honest position is that the silence is real, measurable, and unexplained by anything in our design.

On analysis: Braun and Clarke (2006, 2021) have argued that inter-rater reliability statistics are inappropriate for reflexive thematic analysis, because the analyst’s interpretation is a resource rather than a source of bias. We follow that for the themes. The stance coding is a different kind of exercise, a quantitative content analysis of who is speaking about whom, and there reliability is appropriate. Appendix G reports it, along with its limits.

Appendix B

Data Provenance and Cleaning

The survey ran on Google Forms from 16 to 18 August 2026 and returned 104 responses. The cleaning rule was narrow by design: a value was changed only if it was impossible, and it was changed to missing rather than to a guess. Nothing was capped, winsorised, imputed, or smoothed, and no row was deleted from any file.

Table 2 lists every substantive change. One participant (P001) is excluded from all statistics

because every demographic field was impossible, including an age of 100,000,000 and a department entered as a keyboard mash; that response also arrived about three hours before any other and is almost certainly a test submission. It remains present in every file, flagged rather than deleted. One participant answered the semester question with “Gradeuated”, which is a truthful answer rather than an error; they are retained in full and flagged, and their semester number is missing because there is no number to record.

Two responses were deliberately left alone that a more aggressive cleaner would have removed. One participant wants 20 close friends and another holds 15. Both are high and both are things a student can truthfully report, so they stand. A `straightlining_flag` column records low-variance responding as a diagnostic; it is never used to filter anyone out, and readers can inspect it. Missing data on the core measures were negligible: 98.1% of the analytic sample was complete across all ten.

Table 2

Every substantive change made during data cleaning

ID	Field	Entered	Became	Reason
P001	age	100000000	nan	biologically impossible for an undergraduate
P001	semester	1000th	nan	not a valid semester number (1-16)
P021	semester	Gradeuated	nan	valid answer, not a numbered semester - respondent h
P001	department	bAEdgveargv	nan	not a recognisable BRACU department
P001	gender	homeboy	nan	not a response option offered by the form
P058	friends have	-3	nan	negative friend count is impossible
P058	friends want	-4	nan	negative friend count is impossible
P098	friends want	999	nan	implausible magnitude (>100 close friends)
P001	<entire row>	all identifying fields	excluded from analytic sample	every demographic field impossible or nonsense - not

Note. 19 further entries were case harmonisation on the free-text department field and are omitted. No value was capped, imputed, or smoothed, and no row was deleted from any file.

Appendix C

Measurement

All six scales cleared the conventional $\alpha \geq .70$ threshold, and McDonald’s ω agreed with α to within .02 in every case, which indicates roughly equal item loadings and no single dominant item. The lower bound of the bootstrap interval fell below .70 for anticipated anxiety ($\alpha = .70$, 95% CI [.58, .79]) and for attachment avoidance ($\alpha = .74$, 95% CI [.64, .81]), so results resting on those two are reported with more caution than the rest.

Table 3
Scale statistics and internal consistency

Scale	<i>k</i>	Range	<i>M</i>	<i>SD</i>	α	95% CI	ω	$\bar{r}_{ii}$	Skew
Attachment anxiety	4	1–7	3.48	1.54	.77	[.67, .85]	.78	.46	.21
Attachment avoidance	4	1–7	4.52	1.46	.74	[.64, .81]	.75	.42	-.30
Anticipated anxiety (bids)	6	1–6	2.99	1.10	.70	[.58, .79]	.72	.28	.21
Expected acceptance (bids)	6	1–6	3.74	1.21	.81	[.74, .86]	.82	.42	-.18
Fear of negative evaluation	6	1–5	2.88	1.19	.91	[.87, .94]	.91	.63	-.01
Loneliness (UCLA-3)	3	1–4	2.40	.82	.83	[.75, .88]	.83	.62	-.14

Note. $n = 103$. k is the number of items. The confidence interval for α is a percentile bootstrap over 10,000 resamples. ω is McDonald's omega-total from a one-factor principal-axis solution. $\bar{r}_{ii}$ is the average inter-item correlation.

Table 4
Rotated two-factor solution for the
eight attachment items

Item	Factor 1	Factor 2	h^2
ANX1	.66	.06	.44
ANX2	.79	.10	.64
ANX3	.60	.20	.40
ANX4	.63	.10	.41
AVO1	-.01	.61	.38
AVO2	.26	.70	.55
AVO3	.02	.51	.26
AVO4	.10	.77	.60
% variance	23.7	22.2	

Note. $n = 103$. Principal-axis factoring with varimax rotation. ANX = attachment anxiety items, AVO = attachment avoidance items. KMO = 0.72; Bartlett's test of sphericity $\chi^2(28) = 227.6$, $p < .001$. h^2 is the communality.

The eight attachment items produced a clean two-factor solution (Figure 7), with sampling adequacy and sphericity both acceptable. The scales are measuring two separable things. The problem identified in the main report is therefore not that the avoidance items fail to hang together. They hang together and still fail to predict.

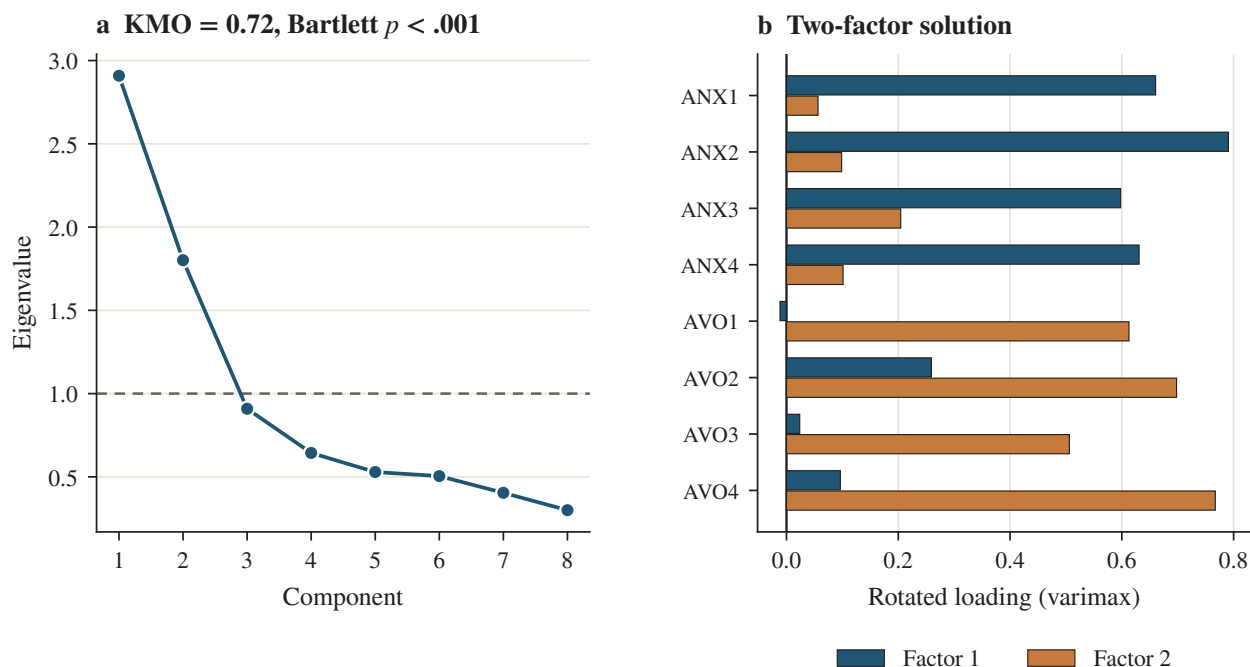


Figure 7
Scree plot and rotated loadings for the eight attachment items.

Appendix D

An Index That Did Not Work

We report this because it failed, and because reporting only what worked would misrepresent how the analysis actually went.

Before looking at the group comparisons we defined a *distress–outcome dissociation score*, D , intended to capture students who pay a social cost larger than the distress they report. It was the difference between a standardised composite of four cost indicators (loneliness, close friends reversed, approach inhibition, conversations initiated reversed) and a standardised composite of two distress indicators (fear of negative evaluation and anticipated anxiety). The prediction was that dismissing students would score highest.

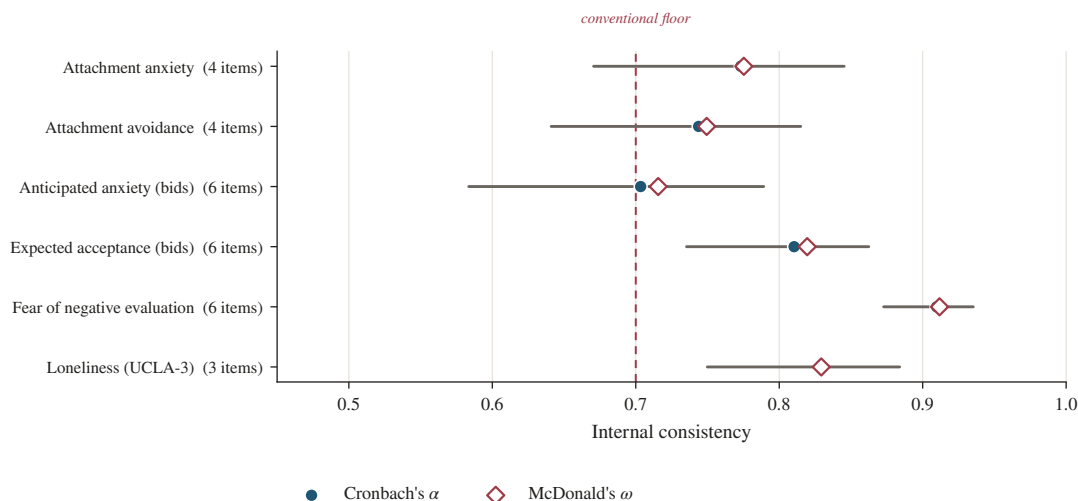


Figure 8

Internal consistency of the six scales, with bootstrap intervals.

The direction was right and the test was not. Dismissing students did score highest ($M = 0.23$) and secure students lowest ($M = -0.21$), but the one-way comparison was not significant, $F(3, 99) = 1.54$, $p = .209$, and D was unrelated to either attachment dimension, $R^2 = .026$, $p = .261$. The reason is visible in the psychometrics: the four-indicator cost composite had an internal consistency of $\alpha = .50$, which is too low to support a difference score. Approach inhibition and conversations initiated simply do not covary with loneliness and friend count tightly enough to be averaged with them.

We then ran a focused version, and we flag it clearly as exploratory rather than planned. Comparing only secure and dismissing students, and using only the two best-measured indicators, the profile interaction was clear: the difference between standardised loneliness and standardised fear of negative evaluation was $+0.49$ for dismissing students and -0.40 for secure students, $t = 3.57$, $p < .001$, Hedges' $g = 0.88$, 95% CI [0.37, 1.39] (Figure 9). Dismissing students report about as little evaluative fear as secure students while living with markedly more loneliness.

We believe this is real, and we are aware that we found it after the planned test failed. It should be treated as a hypothesis for someone else to test on fresh data, not as a result this study establishes.

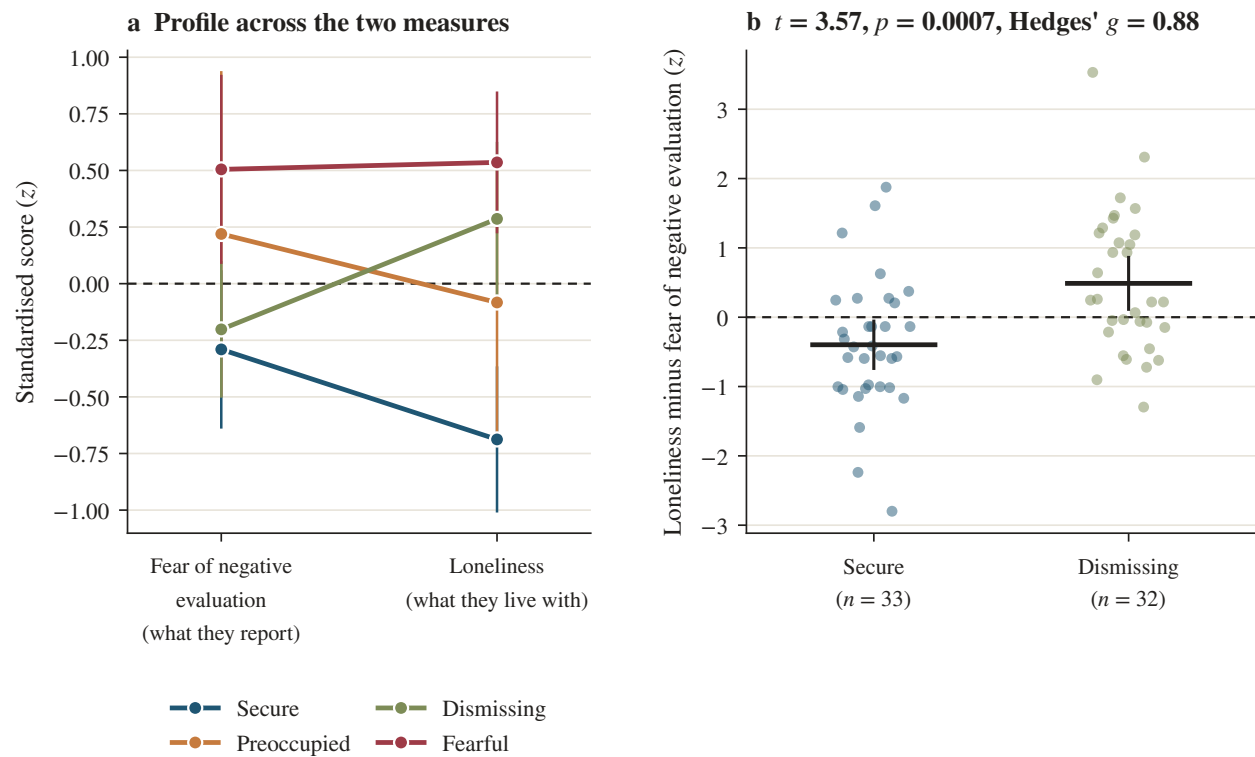


Figure 9

The profile contrast between secure and dismissing students. Panel (a) gives all four styles across the two measures. Panel (b) gives the individual difference scores behind the reported test.

Appendix E

Full Statistical Results

Analyses were run in Python 3 with pandas, NumPy, and SciPy. Every procedure not available in those libraries was implemented directly, including the bootstrap confidence intervals for α , McDonald's ω , principal-axis factoring with varimax rotation, Welch's and Brown–Forsythe tests, the Games–Howell procedure, Pillai's trace, k -means with the adjusted Rand index, Cochran's Q , logistic regression by Newton–Raphson, and the achieved and sensitivity power calculations. The random seed was fixed at 20260827 and all bootstraps used 10,000 resamples. Code is archived (Appendix H) and rerunning it reproduces every number in this report.

Sensitivity. With $n = 103$ and four groups, we had 80% power to detect $f = 0.33$ ($\eta^2 = .10$) in a one-way comparison and $r = .27$ in a correlation. Effects smaller than that are not ruled out by our nulls, and the avoidance comparison in particular had observed power of only .37.

Table 5
One-way comparisons of the four self-selected attachment styles

Outcome	Fisher's F						Welch		K–W	
	F	p	q	η^2	95% CI	ω^2	F	p	H	$1 - \beta$
Attachment anxiety	4.41	.006	.014	.12	[.22, .01]	.09	3.79	.018	11.06	.88
Attachment avoidance	1.36	.259	.287	.04	[.11, .00]	.01	1.22	.314	5.21	.37
Fear of negative evaluation	4.19	.008	.014	.11	[.22, .01]	.09	3.70	.020	10.98	.86
Anticipated anxiety	4.14	.008	.014	.11	[.21, .01]	.08	3.64	.021	10.69	.85
Expected acceptance	1.58	.198	.248	.05	[.12, .00]	.02	1.60	.205	5.18	.42
Loneliness	11.32	< .001	< .001	.26	[.37, .10]	.23	11.37	< .001	25.32	1.00
Close friends held	7.86	< .001	< .001	.19	[.31, .06]	.17	5.85	.002	20.36	.99
Friendship deficit	0.77	.515	.515	.02	[.08, .00]	-.01	0.84	.478	4.25	.22
Approach inhibition	2.60	.057	.081	.07	[.16, .00]	.04	2.92	.047	8.33	.64
Conversations initiated	8.08	< .001	< .001	.20	[.31, .06]	.17	8.08	< .001	18.00	.99

Note. $df = 3, 99$ for Fisher's F except where a case was missing. q is the Benjamini–Hochberg adjusted p across the ten outcomes. The confidence interval for η^2 is derived from the noncentral F . $1 - \beta$ is achieved power at $\alpha = .05$ given the observed effect. K–W is the Kruskal–Wallis rank test.

E.1 Moderation

We tested whether anxiety and avoidance interact, on the theory that the fearful combination might be multiplicative rather than additive. It is not. The interaction term was non-significant for every outcome (all $p > .16$; largest $\beta = -.14$ for conversations initiated), and adding it to the hierarchical

Table 6
Means and standard deviations by self-selected attachment style

Measure	Secure <i>n</i> = 33	Preoccupied <i>n</i> = 11	Dismissing <i>n</i> = 32	Fearful <i>n</i> = 27
Attachment anxiety	3.13 (1.48)	3.75 (1.44)	3.05 (1.26)	4.30 (1.68)
Attachment avoidance	4.30 (1.23)	4.09 (1.31)	4.52 (1.51)	4.95 (1.68)
Fear of negative evaluation	2.53 (1.15)	3.14 (1.26)	2.64 (0.98)	3.48 (1.24)
Anticipated anxiety	2.54 (0.95)	3.02 (0.99)	3.02 (0.96)	3.50 (1.27)
Expected acceptance	4.00 (1.41)	3.29 (1.20)	3.86 (1.17)	3.48 (0.91)
Loneliness	1.84 (0.73)	2.33 (0.73)	2.64 (0.76)	2.84 (0.64)
Close friends held	4.56 (2.96)	2.55 (1.63)	2.69 (1.79)	1.93 (1.73)
Approach inhibition	1.42 (0.90)	1.91 (1.30)	1.94 (1.05)	2.11 (0.97)
Conversations initiated	2.30 (1.36)	0.91 (0.70)	1.62 (1.04)	1.11 (0.80)

Note. Standard deviations in parentheses. Cell *n* varies by one or two for friendship counts because of item non-response.

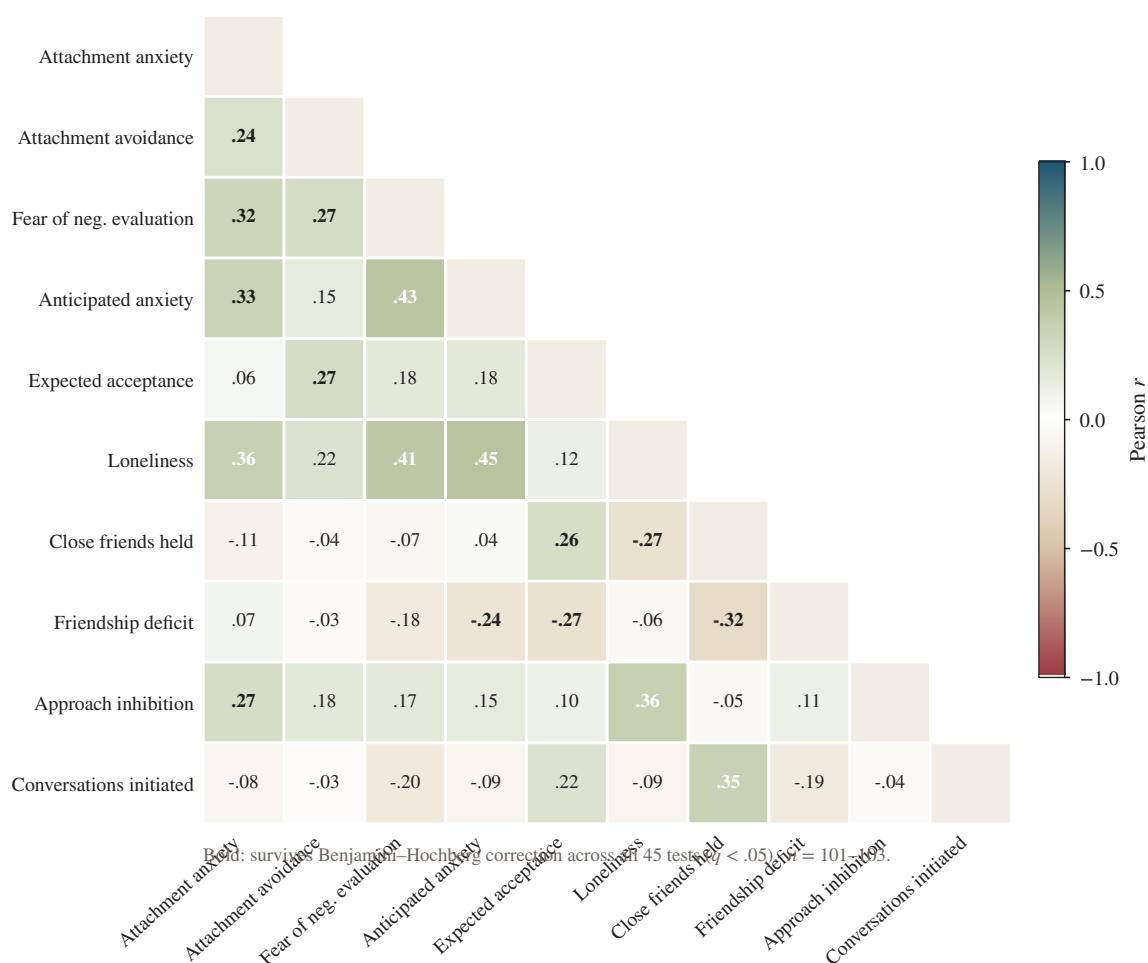


Figure 10
Correlations among the ten core measures.

Table 7
Games–Howell pairwise comparisons

Outcome	Contrast	ΔM	95% CI	d	p
Loneliness	Secure vs. Preoccupied	-.49	[-1.22, .23]	-.68	.246
	Secure vs. Dismissing	-.80	[-1.29, -.31]	-1.07	< .001
	Secure vs. Fearful	-1.00	[-1.47, -.53]	-1.45	< .001
	Preoccupied vs. Dismissing	-.30	[-1.03, .43]	-.40	.652
	Preoccupied vs. Fearful	-.51	[-1.22, .21]	-.76	.224
	Dismissing vs. Fearful	-.20	[-.68, .28]	-.29	.677
Close friends held	Secure vs. Preoccupied	2.02	[.07, 3.96]	.75	.040
	Secure vs. Dismissing	1.88	[.25, 3.50]	.77	.018
	Secure vs. Fearful	2.64	[.99, 4.28]	1.06	< .001
	Preoccupied vs. Dismissing	-.14	[-1.79, 1.51]	-.08	.995
	Preoccupied vs. Fearful	.62	[-1.05, 2.29]	.36	.728
	Dismissing vs. Fearful	.76	[-.45, 1.98]	.43	.355
Conversations initiated	Secure vs. Preoccupied	1.39	[.54, 2.25]	1.13	< .001
	Secure vs. Dismissing	.68	[-.11, 1.47]	.56	.118
	Secure vs. Fearful	1.19	[.44, 1.94]	1.04	< .001
	Preoccupied vs. Dismissing	-.72	[-1.48, .05]	-.74	.074
	Preoccupied vs. Fearful	-.20	[-.93, .53]	-.26	.866
	Dismissing vs. Fearful	.51	[-.12, 1.15]	.55	.152
Attachment anxiety	Secure vs. Preoccupied	-.62	[-2.06, .81]	-.42	.619
	Secure vs. Dismissing	.07	[-.83, .98]	.05	.996
	Secure vs. Fearful	-1.17	[-2.27, -.07]	-.74	.033
	Preoccupied vs. Dismissing	.70	[-.71, 2.10]	.53	.506
	Preoccupied vs. Fearful	-.55	[-2.06, .96]	-.34	.748
	Dismissing vs. Fearful	-1.24	[-2.29, -.19]	-.84	.014
Fear of negative evaluation	Secure vs. Preoccupied	-.61	[-1.84, .63]	-.51	.512
	Secure vs. Dismissing	-.11	[-.80, .59]	-.10	.979
	Secure vs. Fearful	-.95	[-1.77, -.12]	-.79	.019
	Preoccupied vs. Dismissing	.50	[-.71, 1.71]	.48	.637
	Preoccupied vs. Fearful	-.34	[-1.61, .93]	-.27	.873
	Dismissing vs. Fearful	-.84	[-1.62, -.06]	-.76	.031
Anticipated anxiety	Secure vs. Preoccupied	-.47	[-1.45, .50]	-.49	.522
	Secure vs. Dismissing	-.48	[-1.11, .15]	-.50	.191
	Secure vs. Fearful	-.96	[-1.75, -.17]	-.87	.011
	Preoccupied vs. Dismissing	-.01	[-.98, .97]	-.01	1.000
	Preoccupied vs. Fearful	-.48	[-1.55, .58]	-.40	.598
	Dismissing vs. Fearful	-.48	[-1.27, .31]	-.43	.383

Note. Games and Howell's (1976) procedure does not assume equal variances or equal group sizes.

Table 8
Correlations among the ten core measures

	1	2	3	4	5	6	7	8	9	10
1. Attachment anxiety	—									
2. Attachment avoidance	.24	—								
3. Fear of negative evaluation	.32	.27	—							
4. Anticipated anxiety	.33	.15	.43	—						
5. Expected acceptance	.06	.27	.18	.18	—					
6. Loneliness	.36	.22	.41	.45	.12	—				
7. Close friends held	-.11	-.04	-.07	.04	.26	-.27	—			
8. Friendship deficit	.07	-.03	-.18	-.24	-.27	-.06	-.32	—		
9. Approach inhibition	.27	.18	.17	.15	.10	.36	-.05	.11	—	
10. Conversations initiated	-.08	-.03	-.20	-.09	.22	-.09	.35	-.19	-.04	—

Note. $n = 101\text{--}103$. Bold marks correlations surviving Benjamini–Hochberg correction across all 45 tests ($q < .05$); 17 of 45 did so.

Table 9
Each outcome regressed on the two attachment dimensions

Predictor	<i>b</i>	<i>SE</i>	95% CI	β	<i>t</i>	<i>p</i>
<i>Loneliness</i> $R^2 = .15$, $F(2, 100) = 8.68$, $p = < .001$						
Attachment anxiety	.17	.05	[.07, .27]	.32	3.39	< .001
Attachment avoidance	.08	.05	[-.02, .19]	.15	1.55	.124
<i>Close friends held</i> $R^2 = .01$, $F(2, 99) = 0.66$, $p = .521$						
Attachment anxiety	-.17	.16	[-.49, .15]	-.11	-1.06	.290
Attachment avoidance	-.03	.17	[-.36, .31]	-.02	-0.16	.874
<i>Approach inhibition</i> $R^2 = .08$, $F(2, 100) = 4.63$, $p = .012$						
Attachment anxiety	.16	.07	[.03, .29]	.24	2.40	.018
Attachment avoidance	.09	.07	[-.05, .23]	.12	1.25	.215
<i>Conversations initiated</i> $R^2 = .01$, $F(2, 100) = 0.34$, $p = .714$						
Attachment anxiety	-.06	.08	[-.22, .10]	-.08	-0.77	.446
Attachment avoidance	-.01	.08	[-.17, .15]	-.01	-0.11	.912
<i>Fear of negative evaluation</i> $R^2 = .14$, $F(2, 100) = 8.23$, $p = < .001$						
Attachment anxiety	.21	.07	[.06, .35]	.27	2.84	.006
Attachment avoidance	.17	.08	[.01, .32]	.20	2.15	.034
<i>Anticipated anxiety</i> $R^2 = .12$, $F(2, 100) = 6.63$, $p = .002$						
Attachment anxiety	.22	.07	[.09, .36]	.31	3.26	.002
Attachment avoidance	.06	.07	[-.08, .20]	.08	0.82	.415

Note. $n = 101\text{--}103$. Predictors are mean-centred. All variance inflation factors were below 1.1.

Table 10
Bootstrapped mediation models

Path	<i>a</i>	<i>b</i>	<i>c</i>	<i>c'</i>	<i>ab</i>	95% CI	CI excludes 0
Anxiety → FNE → approach inhibition	.25	.08	.18	.16	.020	[-.026, .084]	no
Anxiety → FNE → loneliness	.25	.23	.19	.13	.056	[.013, .106]	yes
Avoidance → conversations → friends	-.02	.71	-.07	-.05	-.016	[-.146, .113]	no
Avoidance → expected acceptance → friends	.23	.60	-.07	-.20	.135	[.038, .264]	yes
FNE → anticipated anxiety → inhibition	.40	.09	.15	.11	.036	[-.047, .128]	no

Note. Percentile bootstrap, 10,000 resamples. FNE = fear of negative evaluation. The avoidance-to-friends model is a suppression case: the total effect is negative while the indirect effect is positive, so it is reported for completeness and not interpreted as mediation.

Table 11
The six friendship bids

Bid	Anticipated anxiety	Expected acceptance	<i>r</i>	<i>p</i>
Sitting together	2.86 (1.74)	3.53 (1.66)	.15	.138
Joining a project team	3.14 (1.71)	3.83 (1.59)	.11	.269
Suggesting lunch	3.60 (1.75)	3.44 (1.63)	-.00	.970
Asking for an explanation	2.24 (1.52)	4.32 (1.78)	.10	.310
Asking about an outing	3.16 (1.91)	3.50 (1.77)	.13	.201
Naming an upset	2.95 (1.73)	3.83 (1.68)	-.02	.871
All six averaged	2.99	3.74	.18	.076

Note. $n = 103$. Both ratings are on 1–6 scales; standard deviations in parentheses. r is the correlation, across students, between how anxious a student expected to feel and how well they expected the bid to go. Across the six bids taken as units the relationship is strong and negative, Spearman $\rho = 0.83$, $p = .042$.

models raised R^2 by no more than .019. We report this as a clean null.

E.2 Empirical clusters against self-selected labels

A k -means solution on standardised anxiety and avoidance, with $k = 4$, recovered clusters that were nearly independent of the prototype students chose: adjusted Rand index = .028, raw agreement 35.0%, $\chi^2(9) = 16.9$, $p = .054$ (Figure 11). Average silhouette width was .38 at $k = 4$ and did not peak there, which is itself consistent with the taxometric literature: no natural break exists to be found. Students were nonetheless internally consistent, choosing the paragraph they had independently rated highest 81.6% of the time, $p < .001$ against chance. They are not answering randomly. They are answering a different question from the one the scales ask.

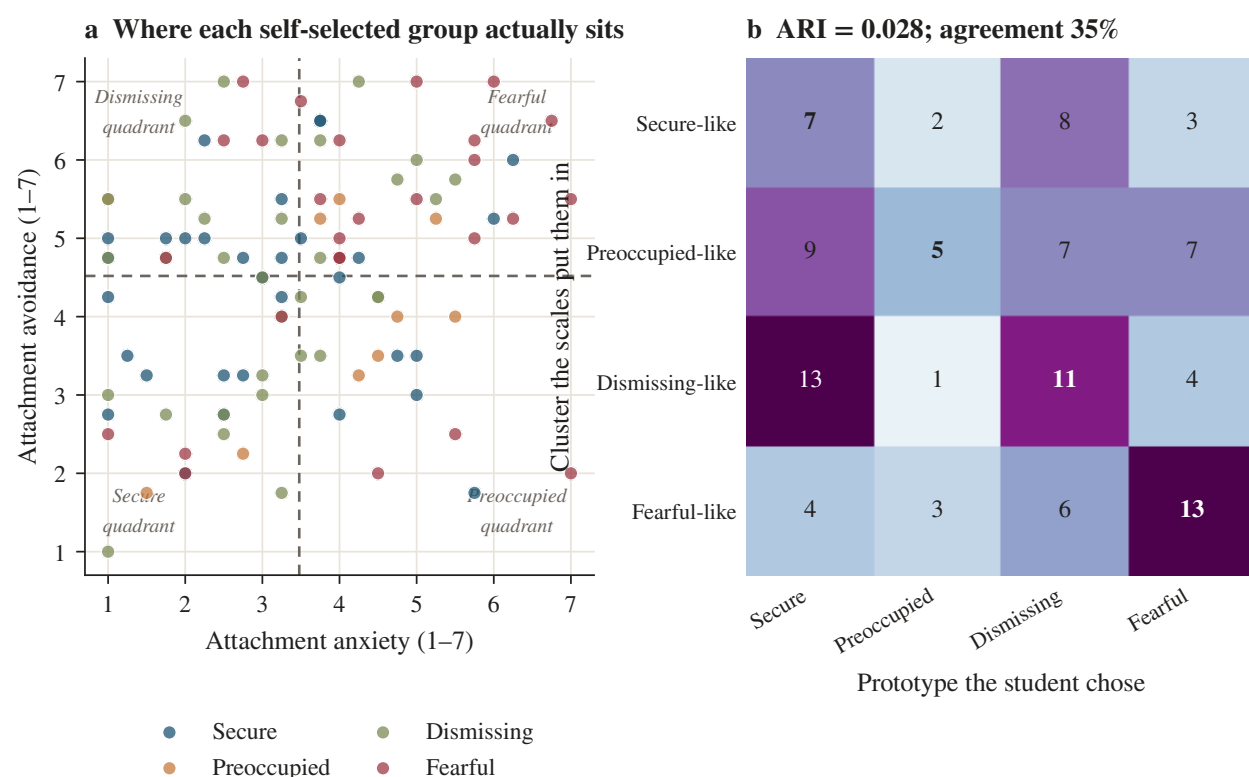


Figure 11

The two-dimensional space, and the correspondence between empirical clusters and self-selected prototypes.

E.3 Household background

Nothing here reached significance. Primary caregiver did not predict attachment anxiety, avoidance, loneliness, or fear of negative evaluation (all $p > .31$), and the number of pre-university moves

or school changes was unrelated to every one of them (all $|\rho| < .09$). Caregiver arrangement was also independent of attachment style, $\chi^2(9) = 10.25$, $p = .330$. Retrospective household variables measured this coarsely, in a sample where 83 of 103 students were raised by both parents, carry no signal. This is a limitation of our instrument rather than evidence against the developmental account.

E.4 Declining the follow-up

Seventy-three of 103 respondents (70.9%, 95% CI [61.1, 79.4]) declined any follow-up interview or discussion. Refusal was flat across attachment styles, $\chi^2(3) = 0.91$, $p = .824$, Cramér's $V = .09$, and a logistic model with anxiety, avoidance, fear of negative evaluation, and loneliness did not predict it either (AUC = .65, model $\chi^2(4) = 5.16$, $p = .272$). We had expected insecure students to opt out at higher rates. They did not. Reticence in this sample is not a property of the insecure; it is the ambient condition, and that null is one of the more interesting things in the study.

Appendix F

Instruments

F.1 Survey

Administered on Google Forms, median completion time under nine minutes. The opening screen carried the consent statement, the voluntary-participation notice, and the Counselling and Wellness Centre contact.

Background. Age; semester; school or department; gender; who raised you for most of your childhood; how often your family moved city or you changed school before university; how many people at BRAC University you would call a close friend, meaning someone you could call at 2 a.m.; how many close friends you would like to have.

Prototypes. The four Bartholomew and Horowitz (1991) paragraphs, each rated 1–7 for self-descriptiveness, followed by a forced choice among them.

Attachment anxiety (1–7). I worry that people close to me will not care about me as much as I care about them. I often worry that the people I am close to will leave me. I need a lot of reassurance that I am cared about. When someone I care about is out of touch for a day, I start assuming something is wrong between us.

Attachment avoidance (1–7). I prefer not to show others how I feel deep down. I find it difficult to depend on other people. I get uncomfortable when someone wants to be very close to me. I would rather solve a problem alone than ask a friend for help.

The six bids. Each rated twice, once for “how anxious or worried would you feel doing this” and once for “how likely is it that the other person says yes or reacts well”, both 1–6. (a) You ask a classmate you have spoken to twice if they want to sit together in the next class. (b) You ask a group in your section if you can join their project team. (c) You message someone from your club and suggest getting lunch. (d) You ask a friend to explain something you did not understand in class. (e) You see friends posting from an outing you were not invited to, and you ask them about it. (f) You tell a close friend that something they did upset you.

Fear of negative evaluation (1–5). I worry about what other people think of me even when I know it makes no difference. I am afraid others will not approve of me. I am afraid that people will find fault with me. When I am talking to someone, I worry about what they may be thinking about me. I often worry that I will say or do the wrong thing. Sometimes I think I am too concerned with what other people think of me.

Behaviour and loneliness. How many times this semester have you started a conversation with a classmate you did not already know (0, 1–2, 3–5, 6–10, more than 10). How often did you want to go up to someone and then decide not to (never to almost every day). How often do you feel that you lack companionship, that you are left out, that you are isolated from others (never, rarely, sometimes, always). Which of these situations do you avoid, ranked: speaking in class, joining a cafeteria group, club or society events, group project meetings, messaging someone first, asking a classmate for help.

F.2 Focus group guide

Sixty minutes, nine participants, one moderator and one note-taker, recorded with written consent. The words *attachment*, *secure*, *anxious*, *avoidant*, and *disorganised* were never spoken in the room, on the reasoning that a labelled participant performs the label.

The vignette, read aloud without comment. “Rafi has been in the same section for five weeks. Every day the same three people sit together near the front, talking and laughing. He has thought about joining them eleven times. Today he opens his phone, types ‘hey, mind if I sit here?’, reads it

twice, and deletes it.”

Questions. Q1 your name, your semester, and if a class got cancelled right now who would you text first, and how did you two actually meet. Q2 one word for walking into a new section in week one. Q3 what is Rafi hearing in his head as he types, in his words. Q4 hands up, who has been Rafi this semester, and what did you do with that feeling. Q5 if he had sent it, what happens. Q6 when Rafi was small and wanted something from an adult in his house, what usually happened. Q7 the person at the back with headphones on who never considers joining: strength, or the same fear in a different shirt. Q8 a close friend has left your message on seen since yesterday and has been online twice. Q9 needing people is weak: who agrees; then, last time you were struggling, how many people knew, on your fingers. Q10 being alone against being left out, and what being left out looks like here on an ordinary Tuesday. Q11 nobody says no out loud, so how do you know you have been rejected, and what happens in your body before you speak to someone new. Q12 is there someone you wish you were closer to, where you are the one keeping the distance. Q13 one sentence you wish somebody had said to you in your first week. Q14 anything said today you disagreed with and did not say at the time.

Q6, Q10, and Q14 produced no usable transcript. Q13 was softened by the moderator in the room into a question about expectations at admission, and it produced more first-person material than any other question in the session.

Appendix G

Focus Group Coding

G.1 The stance scheme

Every turn received one code.

E First-person experiential. The speaker reports something that actually happened to them, or a standing fact about their own life. “In my last semester, I repeated a course, so in that section I had no friends.”

D First-person dispositional. Self-referential but counterfactual or characterological. No instance is given. “I would call one of my friends.” “I’m the type of guy who...”

G Generic or third-person. About Rafi, about “people”, about “humans”, or an abstract principle.
 “From the dawn of civilization, I think people mostly lived alone.”

A Agreement token. Minimal assent, no propositional content. Excluded from the denominator.

Q Peer question. A question put to another participant. Excluded from the denominator.

Turns were also tagged against the six constructs in the moderator guide, and hedging was counted with a fixed lexicon.

G.2 Reliability, and what it is not

One team member coded the corpus by hand against the definitions above. A deterministic rule-based coder was then written from the same definitions and applied to the same 74 turns. Agreement was 64.9% and Cohen’s $\kappa = .47$, 95% CI [.31, .64]. Disagreement concentrated almost entirely on the boundary between D and G, where a generic “I” (“after all, I’m human, I need others”) has to be told apart from a dispositional one. The E category, which carries the paper’s argument, was the most rule-expressible.

This is a transparency check on whether the coding rules are explicit enough to be reapplied mechanically. It is not inter-rater reliability, which would require a second independent human coder, and we did not have one. Readers who want to recode the corpus can: the full coded transcript is archived as a CSV with one row per turn.

G.3 Themes

Four themes were developed reflexively (Braun & Clarke, 2006, 2021), with a fifth that turned out to be the study’s argument.

1. Self-sufficiency as an identity rather than a coping strategy. Independence is not described as something participants do when help is unavailable; it is described as who they are. “I don’t really want to depend that much on someone like that.” One participant set a four-hour rule for himself: if a message about something serious goes unanswered for four hours, he deletes it, because four hours is enough to get out of the state on his own.

2. Rejection is read off the room, never spoken. Asked how they know they have been rejected, the group named body language, tone, vibe, and the environment. Nobody described being told anything. A rejection inferred from atmosphere can never be disconfirmed, which makes the

Table 12
Focus group stance, by moderator question

Question	Demands an instance	Turns	Words	E	D	G	% E
Q1	yes	5	199	0	5	0	0.0
Q2	no	4	20	0	4	0	0.0
Q3	no	4	241	0	0	4	0.0
Q4	yes	3	283	1	0	2	33.3
Q5	no	4	280	0	0	4	0.0
Q7	no	4	395	2	1	1	50.0
Q8	yes	11	644	2	7	2	18.2
Q9	yes	6	306	0	0	6	0.0
Q11	yes	11	236	0	4	7	0.0
Q12	yes	9	237	0	4	5	0.0
Q13	no	10	721	7	2	1	70.0

Note. E = first-person experiential, D = first-person dispositional, G = generic or third-person. Agreement tokens and peer questions are excluded. Q6, Q10, Q14 produced no usable transcript and are absent from the table.

Table 13
Focus group stance, by participant

Participant	Turns	Words	E	D	G	% E	Hedges/100 words
P6	11	899	4	4	3	36.4	1.78
P1	12	689	1	4	7	8.3	2.61
P2	12	566	0	6	6	0.0	1.77
P4	12	337	1	4	5	10.0	2.37
P8	6	298	1	2	3	16.7	3.36
P5	8	257	3	3	2	37.5	2.33
P9	6	253	1	1	3	20.0	1.98
P7	4	182	1	0	3	25.0	2.20
P3	3	100	0	3	0	0.0	0.00

Note. Participants are ordered by words contributed. The proportion of experiential turns did not differ reliably across speakers, $\chi^2(8) = 9.62$, $p = .29$.

inference safe and permanent.

3. The open-credit alibi. The flexible-credit system supplies a structural, non-psychological explanation for thin friendship that costs nothing to give. “Because it’s open credit, I didn’t expect people to stick around.” We think this is partly true and also doing work.

4. The problem is located inside the self, and therefore cannot be discussed. “It’s on us that we think or overthink. It’s our own psychological problem.” “They don’t really know we are debating whether we want them as a friend or not. They are just being themselves. That’s completely my problem.” Once the difficulty is private property, raising it with anyone becomes a category error.

5. Recognition without confession. This is the meta-theme. The room could narrate Rafi’s interior sentence, argue about his childhood in the abstract, and predict his behaviour accurately. Asked to claim any of it, it went quiet or generic. The single clearest moment of disclosure in sixty minutes was four words long and was about the character: “I have been Rafi.”

Appendix H

Ethics, Contributions, and Data Availability

H.1 Ethics

Participation was voluntary throughout and no data were collected before the proposal was approved. Survey respondents read a consent statement before any item and could close the form at any point. Focus group participants gave written consent covering recording and quotation, were told they could pass on any question or stop at any time, and agreed among themselves to confidentiality before the recording began. Names became pseudonyms at transcription; the mapping is held only by the research team and appears in no archived file. Every participant received the contact details of the BRAC University Counselling and Wellness Centre, and the moderator repeated them at the close of the session. We describe patterns and diagnose nobody, and no participant was told which attachment style their answers suggested.

Forty-three respondents gave a name and 22 gave a phone number or email address for follow-up contact. Those fields exist in one identified file that is not archived publicly and is not attached to this report. Everything released publicly is de-identified.

H.2 Author contributions

Md. Monjurul Hasan Bhuiyan designed the survey instrument, built and ran the analysis, and drafted the report. Khan Abdullah co-designed the focus group guide and moderated the session. Azmyeen Nahian kept the field notes during the session and managed recruitment. KM Mahir Labib coded the transcript and cross-checked the stance scheme against the recording. Tasfia Hasan handled consent and transcription and prepared the reference list. Maysha Mahjabin cross-checked the cleaning audit and verified the reported statistics against the analysis output. All six authors read and approved the final version.

H.3 Data availability

The de-identified dataset, the codebook, the cleaning audit log, both instruments, the coded focus group transcript, and all analysis and figure code are openly archived and permanently citable.

Repository: <https://github.com/s0-0han/attachment-friendship-study>

Archived release and DOI: <https://doi.org/10.5281/zenodo.22118219>

Contents: data/ de-identified responses, codebook, audit log, analytic sample; analysis/ the four Python scripts that produce every number and figure; figures/ all eleven figures as PDF and SVG; instruments/ the survey and the focus group guide; fgd/ the coded transcript, one row per turn.

Rerunning `stats_core.py`, `stats_model.py`, `stats_extra.py`, `fgd_coding.py`, and `figures.py` in that order reproduces the entire results section from the raw export. The identified file containing optional names and contact details is deliberately not included.